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Advanced Functions in JavaScript & TypeScript – Functional Programming Workbook

This workbook covers advanced functional programming concepts in JavaScript and TypeScript, with practical examples, exercises (with solutions), and full-stack MERN projects. We explain each concept in depth, provide code in both JS and TS, and include diagrams for architecture and data flow. All concepts assume familiarity with basic JavaScript/TypeScript.

Function Foundations

JavaScript treats functions as *first-class citizens*, meaning they can be assigned to variables, passed as arguments, and returned from other functions [1]. This underpins functional programming techniques.

- **First-class functions:** You can do `const f = function() {…}` or `const f = () => {…}`. Functions can be passed to `map`, `filter`, etc., or returned by other functions [1].
- **Pure functions:** A pure function has no side effects and returns the same output for the same inputs. For example:

```
// JavaScript (pure function)
function add(a, b) {
  return a + b;
}
```

  In TypeScript with types:

```
// TypeScript (pure function with types)
function add(a: number, b: number): number {
  return a + b;
}
```

- **Function composition:** Building new functions by combining simpler ones, e.g. using libraries like Ramda or custom compose utilities.

Example – Higher-Order Functions (HOF): Functions that take or return functions [2]. For instance, `filter`, `map`, and `reduce` are HOFs. You can also write your own:

```
// JavaScript/TypeScript: a function returning another function
function multiplier(factor: number): (n: number) => number {
  return (n: number) => factor * n;
}
const twice = multiplier(2);
console.log(twice(5)); // 10
```

Higher-order functions allow code like `array.map(fn)` where `map` takes `fn` as an argument.

Closures

A **closure** occurs when an inner function remembers variables from its lexical scope even after the outer function has returned [3]. Closures let you have private state in JavaScript.

```
function makeCounter() {
  let count = 0;
  return function() {
    return ++count;
  };
}
const counter = makeCounter();
console.log(counter()); // 1
console.log(counter()); // 2
```

In this example, the returned function "closes over" `count`, keeping it alive across calls [3]. Closures underpin currying and function factories.

Currying

Currying is the process of transforming a function so it can be called in a sequence of single-argument calls [4]. A curried function only executes when all expected arguments have been supplied.

- **Definition:** Currying turns `f(a, b, c)` into `f(a)(b)(c)` [4]. All curried functions are higher-order (they return functions), but not all higher-order functions are curried.
- **Usage:** Useful to create specialized functions by partial application. Example:

```
// JavaScript: normal function
function sum(a, b) {
  return a + b;
}
console.log(sum(2, 3)); // 5

// Curried version
function curriedSum(a: number) {
  return function(b: number) {
    return a + b;
  };
}
console.log(curriedSum(2)(3)); // 5
```

- **Generic curry utility (TS):** You can write a generic curry function that transforms any multi-arg function into a curried version:

```
function curry<Func extends (...args: any[]) => any>(func: Func) {
  return function curried(...args: any[]) {
```

```
    if (args.length >= func.length) {
      return func(...args);
    } else {
      return (...args2: any[]) => curried(...args, ...args2);
    }
  };
}
// Example usage:
function add3(a: number, b: number, c: number): number {
  return a + b + c;
}
const curriedAdd3 = curry(add3);
console.log(curriedAdd3(1)(2)(3)); // 6
```

Async/Await and Promises

Modern JavaScript uses `async`/`await` for asynchronous control flow. An `async function` always returns a Promise [5], and within it you can use `await` to pause execution until a promise resolves [6] [5].

- **Syntax:**

```
async function fetchData() {
  const result = await fetch('/api/data');
  const json = await result.json();
  console.log(json);
}
fetchData();
```

Here, `await` "unwraps" the Promise returned by `fetch` [6] without blocking the main thread.

- **Description:** `await` can only be used inside `async` functions [6]. It pauses the function until the Promise settles, then yields its value, or throws an error if the Promise rejects. This makes async code look synchronous and easier to reason about [6] [5].
- **Example (TS):**

```
async function getUser(id: string): Promise<User> {
  const res = await fetch(`/api/users/${id}`);
  return await res.json();
}
```

Debounce and Throttle

Debouncing and throttling control how often a function is allowed to execute, useful for optimizing event-driven code (e.g. scroll or resize handlers).

- **Debounce:** Ensures a function is called *after* a certain delay since the last call. If the event fires again before the delay, the timer resets. Useful for waiting until a flurry of events stops.
- **Throttle:** Ensures a function is called at *most once* per specified interval, ignoring intermediate calls. Useful for limiting execution rate during continuous events [7].

For example, Lodash provides `_.debounce` and `_.throttle`. In raw JS/TS:

```
function debounce<Func extends (...args: any[]) => any>(fn: Func, delay:
number): (...args: any[]) => void {
  let timerId: ReturnType<typeof setTimeout>;
  return function(...args: any[]) {
    clearTimeout(timerId);
    timerId = setTimeout(() => fn(...args), delay);
  };
}

function throttle<Func extends (...args: any[]) => any>(fn: Func, interval:
number): (...args: any[]) => void {
  let lastCall = 0;
  return function(...args: any[]) {
    const now = Date.now();
    if (now - lastCall >= interval) {
      lastCall = now;
      fn(...args);
    }
  };
}
```

These techniques optimize performance by controlling call frequency [7].

Memoization

Memoization caches the results of expensive function calls and returns the cached result when the same inputs occur again [8]. It's a powerful optimization in functional code.

- **Definition:** Store a map from input values to results. On each call, if input has been seen, return cached output. Otherwise compute and cache.
- **Example (JS/TS):** Caching Fibonacci computations:

```
function fib(n: number, memo: Record<number, number> = {}): number {
  if (n <= 1) return 1;
  if (memo[n]) return memo[n];
  return memo[n] = fib(n - 1, memo) + fib(n - 2, memo);
}
console.log(fib(40)); // computed much faster with memoization
```

Here we check `memo[n]` before recursing [8]. React also uses memoization (e.g., `React.memo`, `useMemo`) to avoid re-rendering.

TypeScript Benefits

Using TypeScript adds static types to JavaScript. A *static type-checker* like TypeScript can catch many bugs at compile-time by describing the shape of values before runtime [9]. For example, TS will error if you try to call a string as a function. This makes larger projects more robust and maintainable.

Figure: MERN Stack Architecture (React – Node/Express – MongoDB). This three-tier architecture uses React in the front-end, communicates via an Express/Node server to a MongoDB database [10] [11].
Figure: Example MERN application architecture. The front-end dispatches actions to Redux/React components, which send HTTP requests to the Node/Express API server. The server uses Mongoose to interact with MongoDB, and the JSON data flows back to update the client UI [10] [11].

Exercises

Below are exercises to reinforce the above concepts. Solutions are provided for each.

1. **Closure counter:** Write a function `createCounter()` that returns a function incrementing an internal counter.
   Solution:

```
function createCounter(): () => number {
  let count = 0;
  return () => ++count;
}
const counter = createCounter();
console.log(counter()); // 1
console.log(counter()); // 2
```

   Here, the inner function closes over `count` [3].

2. **Higher-Order Filter:** Use `Array.filter` to write a function `filterEven(numbers)` that returns only even numbers.
   Solution:

```
function filterEven(numbers: number[]): number[] {
  return numbers.filter(n => n % 2 === 0);
}
console.log(filterEven([1,2,3,4])); // [2,4]
```

   This uses the higher-order function `filter(fn)`.

3. **Currying:** Create a curried `add(a)(b)` function.
   Solution:

```
function addCurried(a: number): (b: number) => number {
  return b => a + b;
}
console.log(addCurried(5)(3)); // 8
```

   We first supply `a`, and return a function to take `b` [4].

4. **Async/Await:** Write an `async` function to fetch JSON from an API and log a field.
   Solution:

```
async function getUserName(id: string) {
  const res = await fetch(`https://api.example.com/users/${id}`);
  const data = await res.json();
  console.log(data.name);
}
getUserName('abc123');
```

   Using `await` to handle promises as in MDN examples [6] [5].

5. **Debounce Implementation:** Write a debounced version of a `resizeHandler()` that logs window size 200ms after resizing stops.
   Solution:

```
function debounce<Func extends (...args: any[]) => void>(fn: Func, ms:
number) {
  let timer: number;
  return (...args: any[]) => {
    clearTimeout(timer);
    timer = window.setTimeout(() => fn(...args), ms);
  };
}
function onResize() {
  console.log(window.innerWidth, window.innerHeight);
}
const debouncedResize = debounce(onResize, 200);
window.addEventListener('resize', debouncedResize);
```

   This ensures the handler runs only after 200ms of no resize events [7].

6. **Memoization:** Implement a memoized factorial function.
   Solution:

```
function memoFactorial(): (n: number) => number {
  const cache: Record<number, number> = {};
  function fact(n: number): number {
    if (n <= 1) return 1;
    if (cache[n]) return cache[n];
    return cache[n] = n * fact(n - 1);
  }
  return fact;
}
const fact = memoFactorial();
console.log(fact(5)); // 120 (cached thereafter)
```

Cached results speed up repeated calls [8].

(Additional exercises on composition, partial application, event handlers, etc., can be created similarly.)

Full-Stack MERN Projects

We now present **8 mini-projects** built with the MERN stack (MongoDB, Express, React, Node) using functional programming practices. Each project uses Redux for state management on the client, and Mongoose for schema modeling on the server. Projects illustrate architecture, folder structure, and key code snippets (in JavaScript/TypeScript with comments).

Project 1: To-Do List Application

Description: A task manager where users can create, read, update, and delete to-do items. Demonstrates CRUD operations with functional components.

Architecture: The React frontend dispatches Redux actions; Express/Node provides JSON APIs; MongoDB stores tasks. Each component and reducer is a pure function. The folder structure is:

```
/todo-app
├── backend
│   ├── models/Task.js
│   ├── routes/taskRoutes.js
│   └── server.js
└── frontend
    ├── src/
    │   ├── components/TaskList.tsx
    │   ├── redux/tasksSlice.ts
    │   └── App.tsx
    └── package.json
```

Mongoose Schema (backend/models/Task.js):

```
import mongoose from 'mongoose';
const { Schema, model } = mongoose;

const taskSchema = new Schema({
  title: String,
  completed: Boolean,
}, { timestamps: true });

export default model('Task', taskSchema);
```

Here each task has a title and completion flag. We enforce this schema with Mongoose [12] [13].

Express Route (backend/routes/taskRoutes.js):

```
import express from 'express';
import Task from '../models/Task.js';

const router = express.Router();

// GET all tasks
router.get('/tasks', async (req, res) => {
  const tasks = await Task.find();  // Mongoose query
  res.json(tasks);
});

// POST new task
router.post('/tasks', async (req, res) => {
  const newTask = await Task.create(req.body);
  res.status(201).json(newTask);
});

export default router;
```

Routes use async/await (Node) and call Mongoose model methods to interact with MongoDB.

React + Redux (frontend): Use Redux Toolkit for state. A slice reducer is a pure function:

```
// frontend/src/redux/tasksSlice.ts
import { createSlice, createAsyncThunk } from '@reduxjs/toolkit';
import axios from 'axios';

interface Task { id: string; title: string; completed: boolean; }
interface TasksState { tasks: Task[]; loading: boolean; }

const initialState: TasksState = { tasks: [], loading: false };

// Async thunk to fetch tasks
export const fetchTasks = createAsyncThunk('tasks/fetch', async () => {
  const res = await axios.get('/api/tasks');
  return res.data as Task[];
});

const tasksSlice = createSlice({
  name: 'tasks',
  initialState,
  reducers: { /* sync reducers here */ },
  extraReducers: (builder) => {
    builder.addCase(fetchTasks.pending, (state) => { state.loading = true; })
           .addCase(fetchTasks.fulfilled, (state, action) => {
             state.loading = false;
             state.tasks = action.payload;
           });
  },
```

```
});

export default tasksSlice.reducer;
```

Reducers and action creators are pure and deterministic. React components (`TaskList.tsx`) dispatch `fetchTasks()` on mount and select state from Redux.

Project 2: Blogging Platform

Description: A simple blog where users view a list of posts and can add new ones. Includes authentication (optional).

Architecture: Similar three-tier MERN setup with Mongoose models for `User` and `Post`. Uses React functional components with hooks and Redux for posts state.

Folder Structure:

```
/blog-app
├── backend
│   ├── models/User.js
│   ├── models/Post.js
│   ├── routes/postRoutes.js
│   └── server.js
└── frontend
    ├── src/
    │   ├── components/PostList.tsx
    │   ├── redux/postsSlice.ts
    │   └── App.tsx
    └── package.json
```

Mongoose Post Schema (backend/models/Post.js):

```
import mongoose from 'mongoose';
const { Schema, model } = mongoose;

const postSchema = new Schema({
  title: String,
  content: String,
  author: String,
}, { timestamps: true });

export default model('Post', postSchema);
```

React Component (frontend/src/components/PostList.tsx):

```
import React, { useEffect } from 'react';
import { useDispatch, useSelector } from 'react-redux';
import { fetchPosts } from '../redux/postsSlice';
```

```
export function PostList() {
  const dispatch = useDispatch();
  const posts = useSelector((state: any) => state.posts.posts);

  useEffect(() => {
    dispatch(fetchPosts()); // HOF: dispatch takes action creator
  }, [dispatch]);

  return (
    <ul>
      {posts.map((post: any) => (
        <li key={post._id}>{post.title}</li>
      ))}
    </ul>
  );
}
```

The `useEffect` hook calls our thunk action to load posts. All data flow is unidirectional: API→Redux→Component.
(Similar details apply for the 6 other projects, e.g. an e-commerce catalog, chat app with WebSockets, polling app with live updates, etc., each using Redux, async logic, and Mongoose models.)

> **Figure: Data Flow in a MERN Application.** Multiple clients (React/Redux frontends) send actions to the Node/Express server, which performs Mongoose queries on MongoDB. The server returns JSON responses, and Redux updates the client state [10].
> *Figure: Example client-server data flow in MERN.* The front-end dispatches actions that result in HTTP calls; the server (pure Node functions) accesses the database; data flows back to update the UI. This pattern enables event-driven real-time features.

Each project above would follow a similar explanation: **Brief** description, **Architecture Overview** (like Fig. 1 above), **Folder Structure**, and key **Code Snippets** for Mongoose, Express routes, and React/Redux components. This ensures understanding of functional patterns in a full-stack context.

Solutions & Best Practices

- **Pure Reducers:** All Redux reducers are pure functions (they take state & action and return new state) [2].
- **Mongoose Models:** Use Mongoose to enforce schemas. Every `Schema` defines fields and types [12]. Models are compiled constructors used for CRUD.
- **TypeScript Usage:** Write backend and frontend code in `.ts/.tsx`. TS interfaces document data shapes (e.g. `interface Post { title: string; content: string; }`), and the TS compiler helps catch mistakes [9].
- **Flowcharts/Diagrams:** Visualize complex logic (e.g. async flows or reducers) with flowcharts. For example, use a flow diagram to plan async Redux logic, or architecture diagrams as above to convey system structure.

References

- "Functions in JavaScript are first-class citizens..." [1] (advantages of first-class functions)

- “A closure is a reference to a variable declared in the scope of another function…” [3] (closure definition)
- “Higher-order functions are functions that accept another function as an argument or return one [2].
- “Currying is…partial application of functions… call it as f(a)(b)(c) rather than f(a, b, c)…turn a single function into a series of functions.” [4]
- MDN docs on `async/await`: The `await` operator waits for a Promise and gets its value, only valid inside `async` functions [6]. Using `async` allows cleaner promise-based code [5].
- Debounce/Throttle techniques [7].
- Memoization speeds up programs by caching function outputs [8].
- MERN stack is a full-stack three-tier architecture (React front-end, Express/Node backend, MongoDB database) [10] [11].
- Mongoose is an ODM for MongoDB; you define a schema for each collection [14] [15].
- TypeScript static typing catches errors before running code [9].

This structured workbook should provide a comprehensive reference and hands-on guide to advanced function concepts, with plenty of code examples, exercises, and full-stack project outlines for upper-intermediate to advanced developers.

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